

DAY 04 – TCP/IP & OSI

Networking Assignment Report

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Assignment Code	DAY04-TCPIP-OSI
Total Marks	20
Operating System	Ubuntu Linux

Ubuntu Linux Note: The assignment specifies Windows commands such as tracert. This practical was performed on Ubuntu Linux, so the Linux equivalent traceroute was used.

OSI/TCP-IP THEORY

1. What is a network protocol?

A network protocol is a set of rules that define how devices communicate and transfer data over a network. Examples of protocols are TCP, IP, HTTP, HTTPS, DNS, and Ethernet.

2. What is the OSI Model?

The Open Systems Interconnection (OSI) Model is a seven-layered conceptual framework used to understand how data is transferred between network devices.

3. Name all seven OSI layers in correct order.

OSI Layers (Top to Bottom): Application, Presentation, Session, Transport, Network, Data Link, and Physical.

4. What is the purpose of the Application Layer?

The Application Layer is responsible for providing network services to applications and users. Some protocols of importance are HTTP, HTTPS, DNS, FTP, and SMTP.

5. What is the purpose of the Transport Layer?

The Transport Layer provides end-to-end communication between applications using segmentation, reliability, flow control, and port numbers. Two key protocols are TCP, which provides reliable delivery, and UDP, which is connectionless.

6. What is the purpose of the Network Layer?

The Network Layer is responsible for logical addressing and routing of packets between networks. IP addresses and routers primarily operate on this layer.

7. What is the purpose of the Data Link Layer?

The Data Link Layer provides node-to-node transmission over a local network. This layer uses frames and MAC addresses to deliver data with error detection and media access control functions.

8. What is the purpose of the Physical Layer?

The Physical Layer is responsible for transmission of raw bits over copper, fiber, or radio waves media.

9. What is the TCP/IP Model?

The TCP/IP Model is a practical networking model used as the basis for the Internet. The model's common layers are Application, Transport, Internet, and Network Access.

10. Explain the difference between OSI and TCP/IP models.

The OSI Model is a conceptual model that comprises seven conceptual layers, whereas the TCP/IP suite is more of a practical protocol model with four commonly used layers. Additionally, some layers in the OSI Model are combined into one layer in the TCP/IP suite.

11. What is encapsulation?

Encapsulation refers to the process of attaching control information such as headers and trailers to data as it travels down the networking layers.

12. What is decapsulation?

Decapsulation is a process of decapsulating data by each layer in the receiver device to extract relevant control information.

13. Which layer uses IP addresses?

The Network layer (OSI Layer 3) is responsible for utilizing IP addresses for logical addressing and routing of packets.

14. Which layer uses MAC addresses?

The Data Link layer (OSI Layer 2) is responsible for utilizing MAC addresses for node-to-node communication.

15. Which layer uses TCP and UDP?

TCP and UDP are communication protocols that operate in the Transport Layer (OSI Layer 4).

16. Which layer includes HTTP and HTTPS?

HTTP and HTTPS are Application-layer protocols (OSI Layer 7).

17. Why should cybersecurity professionals understand networking layers?

Networking layers provide invaluable information for analyzing traffic, understanding protocols, locating attacks, troubleshooting incidents, configuring defenses, and detecting suspicious traffic.

18. Explain how network data travels from a browser to a server.

A browser will perform DNS to locate the IP address of a domain and create an HTTP/HTTPS request to the server. The data request will then traverse through the Transport, Network, Data Link, and Physical layers to reach the destination server. Routers will pass the packet of data with an IP address towards the destination server. Finally, the data is decapsulated, processed by the web application, and a response is sent back to the browser.

PRACTICAL

19. Your resolved IP address for google.com (nslookup)

The nslookup output returned these IPv4 addresses: 192.178.158.139, 192.178.158.138, 192.178.158.101, 192.178.158.102, 192.178.158.100, and 192.178.158.113. The traceroute output used 192.178.158.102 as the destination.

20. Number of traceroute hops to google.com

The traceroute output reached the destination at hop 30. Therefore, the observed path contained 30 hops. Some intermediate hops showed '*' because they did not return a traceroute response.

SCREENSHOTS

The assignment requests ping google.com, tracert google.com, and nslookup google.com screenshots. Since Ubuntu Linux was used, traceroute was used instead of Windows tracert.

```
md-hamza-khan@md-hamza-khan-HP-ZBook:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.158.139
Name:   google.com
Address: 192.178.158.138
Name:   google.com
Address: 192.178.158.101
Name:   google.com
Address: 192.178.158.102
Name:   google.com
Address: 192.178.158.100
Name:   google.com
Address: 192.178.158.113
Name:   google.com
Address: 2404:6800:4013:804::64
Name:   google.com
Address: 2404:6800:4013:804::65
Name:   google.com
Address: 2404:6800:4013:804::66
Name:   google.com
Address: 2404:6800:4013:804::71
```

Figure 1 – nslookup google.com

```

md-hamza-khan@md-hamza-khan-HP-ZBook:~$ traceroute google.com
traceroute to google.com (192.178.158.102), 30 hops max, 60 byte packets
 1 _gateway (10.177.40.82) 5.765 ms 5.759 ms 5.834 ms
 2 * * *
 3 255.0.0.2 (255.0.0.2) 81.282 ms 81.649 ms 81.650 ms
 4 255.0.0.3 (255.0.0.3) 81.178 ms 81.142 ms 79.386 ms
 5 192.168.227.99 (192.168.227.99) 81.072 ms 192.168.227.98 (192.168.227.98) 77.666 ms 79.475 ms
 6 172.25.86.126 (172.25.86.126) 79.695 ms 32.913 ms 172.25.119.230 (172.25.119.230) 32.876 ms
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * *
13 * 72.14.195.122 (72.14.195.122) 88.387 ms *
14 142.251.66.185 (142.251.66.185) 87.298 ms 142.251.71.193 (142.251.71.193) 90.506 ms 142.251.66.185 (142.251.66.185) 89.182 ms
15 64.233.174.17 (64.233.174.17) 88.946 ms 142.251.70.211 (142.251.70.211) 88.381 ms 142.251.230.70 (142.251.230.70) 86.915 ms
16 172.253.71.132 (172.253.71.132) 67.049 ms 172.253.73.230 (172.253.73.230) 68.149 ms 192.178.125.8 (192.178.125.8) 68.101 ms
17 172.253.73.230 (172.253.73.230) 68.050 ms 192.178.125.10 (192.178.125.10) 75.456 ms 192.178.252.111 (192.178.252.111) 83.988 ms
18 * * 142.251.227.239 (142.251.227.239) 90.563 ms
19 * 192.178.251.245 (192.178.251.245) 82.254 ms 80.349 ms
20 192.178.252.85 (192.178.252.85) 81.415 ms 192.178.252.81 (192.178.252.81) 80.556 ms *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * lcdels-in-f102.1e100.net (192.178.158.102) 90.424 ms

```

Figure 2 – traceroute google.com

```
md-namza-khan@md-namza-khan-HP-ZBOOK:~$ ping google.com
PING google.com (2404:6800:4013:804::71) 56 data bytes
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=1 ttl=111 time=40.4 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=2 ttl=111 time=57.8 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=3 ttl=111 time=75.7 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=4 ttl=111 time=56.9 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=5 ttl=111 time=48.5 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=6 ttl=111 time=34.2 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=7 ttl=111 time=30.6 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=8 ttl=111 time=38.3 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=9 ttl=111 time=52.8 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=10 ttl=111 time=46.8 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=11 ttl=111 time=35.5 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=12 ttl=111 time=57.8 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=13 ttl=111 time=49.3 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=14 ttl=111 time=46.5 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=15 ttl=111 time=45.6 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=16 ttl=111 time=45.7 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=17 ttl=111 time=71.6 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=18 ttl=111 time=48.5 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=19 ttl=111 time=39.3 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=20 ttl=111 time=41.7 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=21 ttl=111 time=69.4 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=22 ttl=111 time=59.7 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=23 ttl=111 time=60.3 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=24 ttl=111 time=34.5 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=25 ttl=111 time=58.7 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=26 ttl=111 time=68.4 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=27 ttl=111 time=58.9 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=28 ttl=111 time=50.7 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=29 ttl=111 time=38.5 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=30 ttl=111 time=39.0 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=31 ttl=111 time=52.0 ms
64 bytes from nw-in-f113.1e100.net (2404:6800:4013:804::71): icmp_seq=32 ttl=111 time=40.4 ms
```

Figure 3 – ping google.com

COMMANDS USED

nslookup google.com

tracert google.com

ping google.com

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Platform: Ubuntu Linux